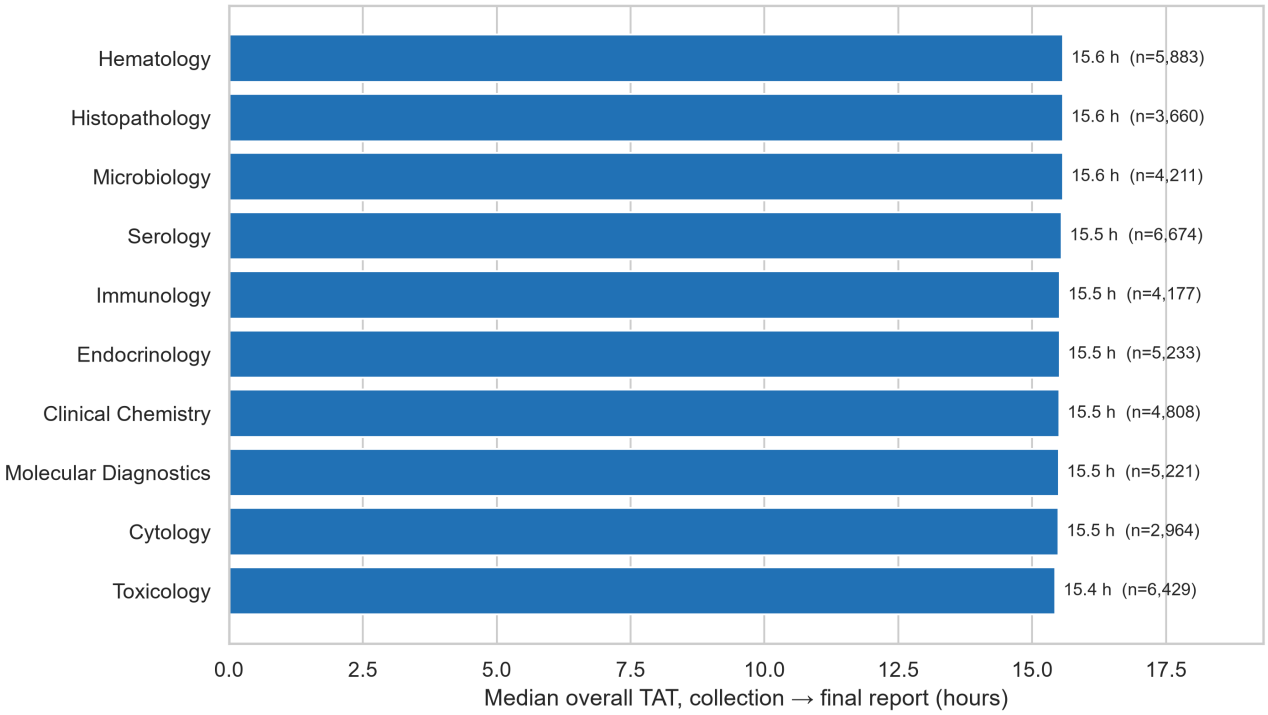
Overall TAT by Department

Median end-to-end turnaround time (collection to final report) for each of the 10 departments, ranked from slowest to fastest.



Key insights

- All 10 departments sit within 15.4 to 15.6 median hours - a spread of only about 9 minutes.
- Hematology is the slowest department (15.58 h); Toxicology the fastest (15.43 h).
- No department is an outlier - the delay is shared by the whole laboratory.

Potential bottleneck

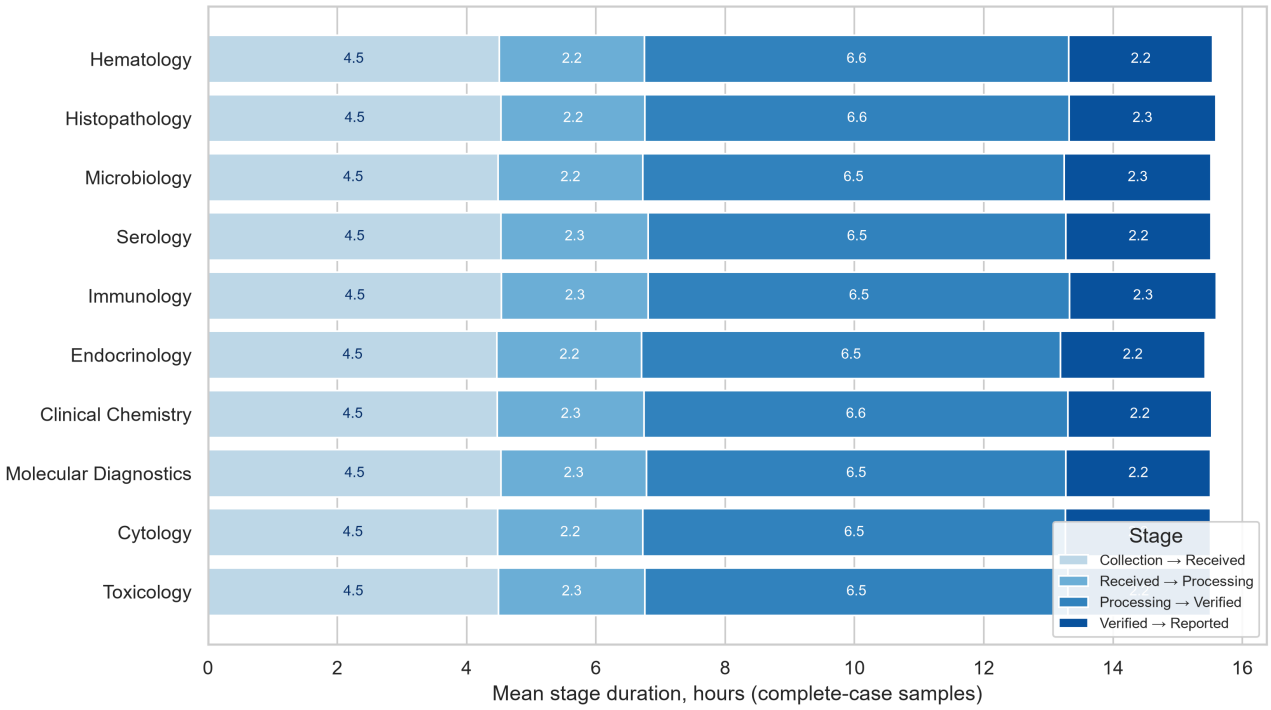
None at department level: slow turnaround is a laboratory-wide process property, not one department's problem.

Suggested action

Do not chase individual departments. Target the slow workflow stage process-wide (see the stage-composition chart on the next page).

Department TAT Composition by Workflow Stage

How each department's overall TAT splits across the four workflow stages; the thickest segment is that department's dominant delay.



Key insights

- Processing -> Verified is the largest segment in every department: about 6.5 h, 42% of mean TAT.
- Collection -> Received adds about 4.5 h (29%); the other two stages are minor at roughly 2.2-2.3 h (14-16%).
- The composition is almost identical across all departments.

Potential bottleneck

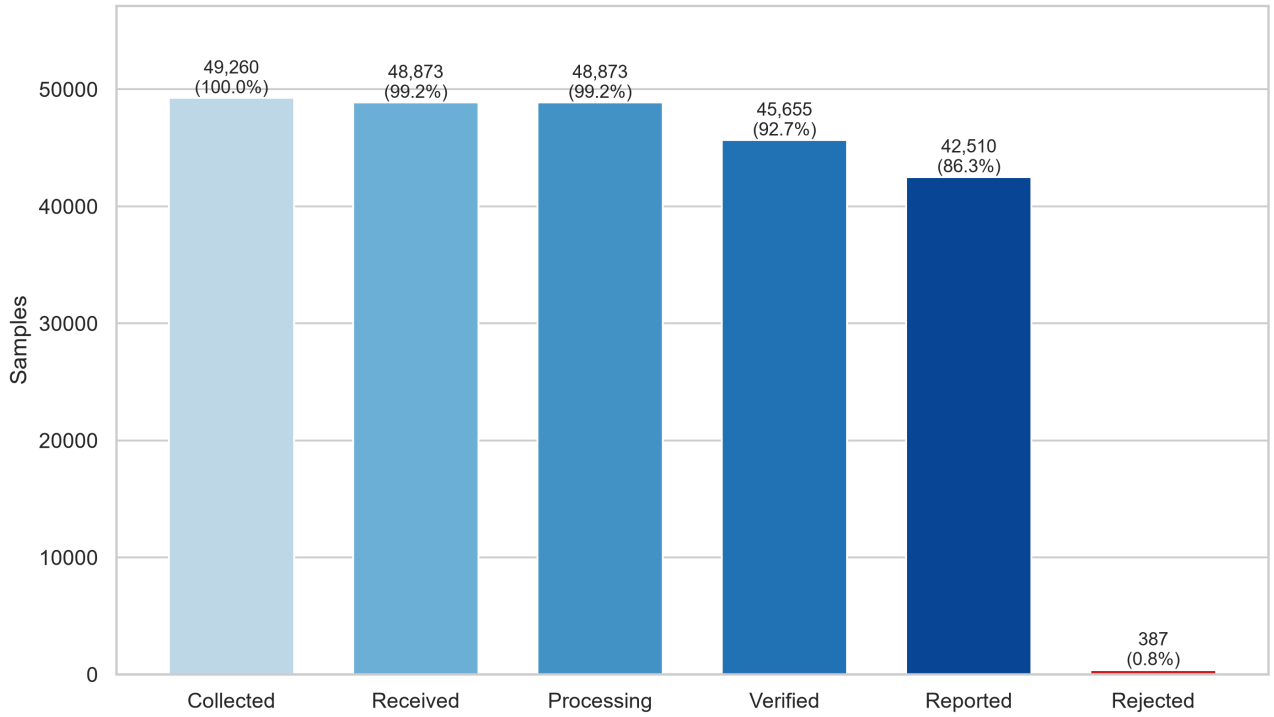
The processing/verification bench (Processing -> Verified) is the single heaviest stage laboratory-wide.

Suggested action

Concentrate improvement on this stage: coverage at verified-time, automation of checks, and review of queuing before verification.

Samples Reaching Each Workflow State

How many samples progress through collected, received, processing, verified and reported - and how many exit as rejected.



Key insights

- 49,260 samples were collected; 42,510 (86.3%) reached a final report.
- 387 samples (0.79%) rejected before processing started.
- 6,750 samples (13.7%) were neither reported nor rejected - open work at the data snapshot.

Potential bottleneck

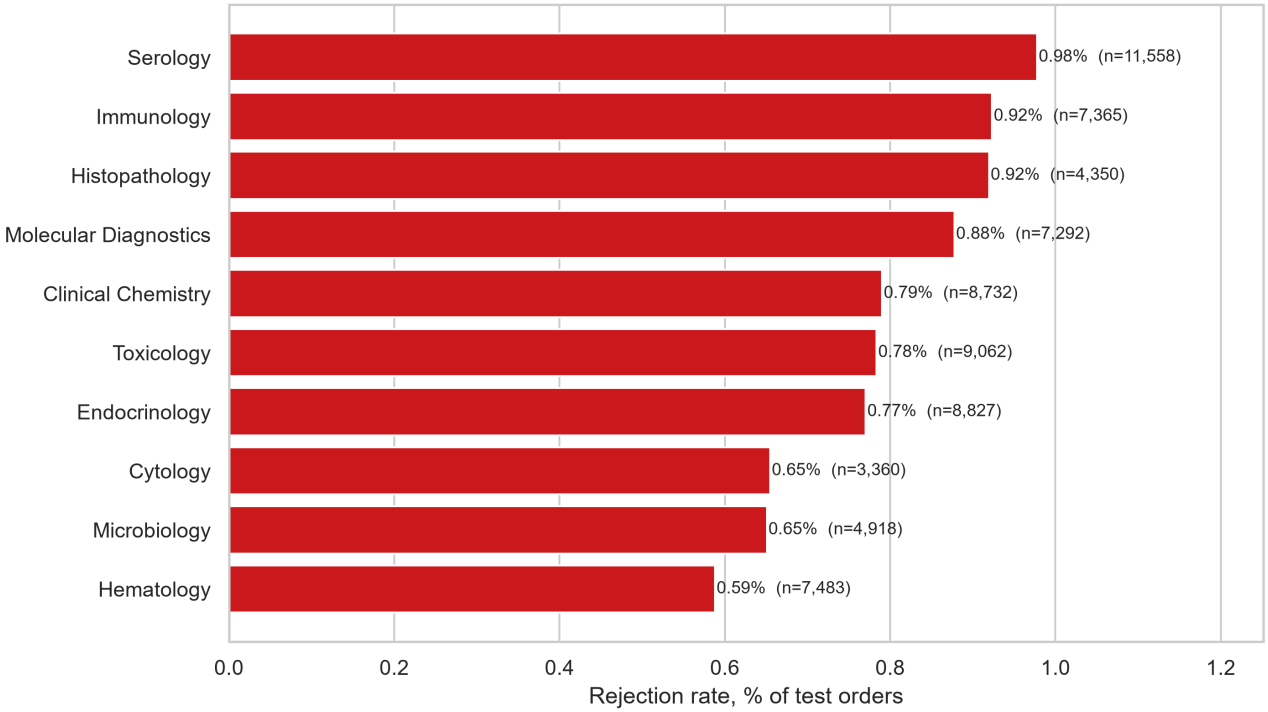
Attrition concentrates after processing: roughly 6.2k samples sit between verified and reported, plus the rejections.

Suggested action

Review the unfinished backlog weekly and chase its oldest samples; keep the reported share moving towards 100%.

Rejection Rate by Department

Rejections as a share of test orders per department - the department view of collection quality.



Key insights

- Serology records the highest rate at 0.98% (113 of 11,558 orders); Immunology and Histopathology follow at 0.92%.
- Hematology is the cleanest at 0.59%.
- Every department keeps rejections below 1% of its test orders.

Potential bottleneck

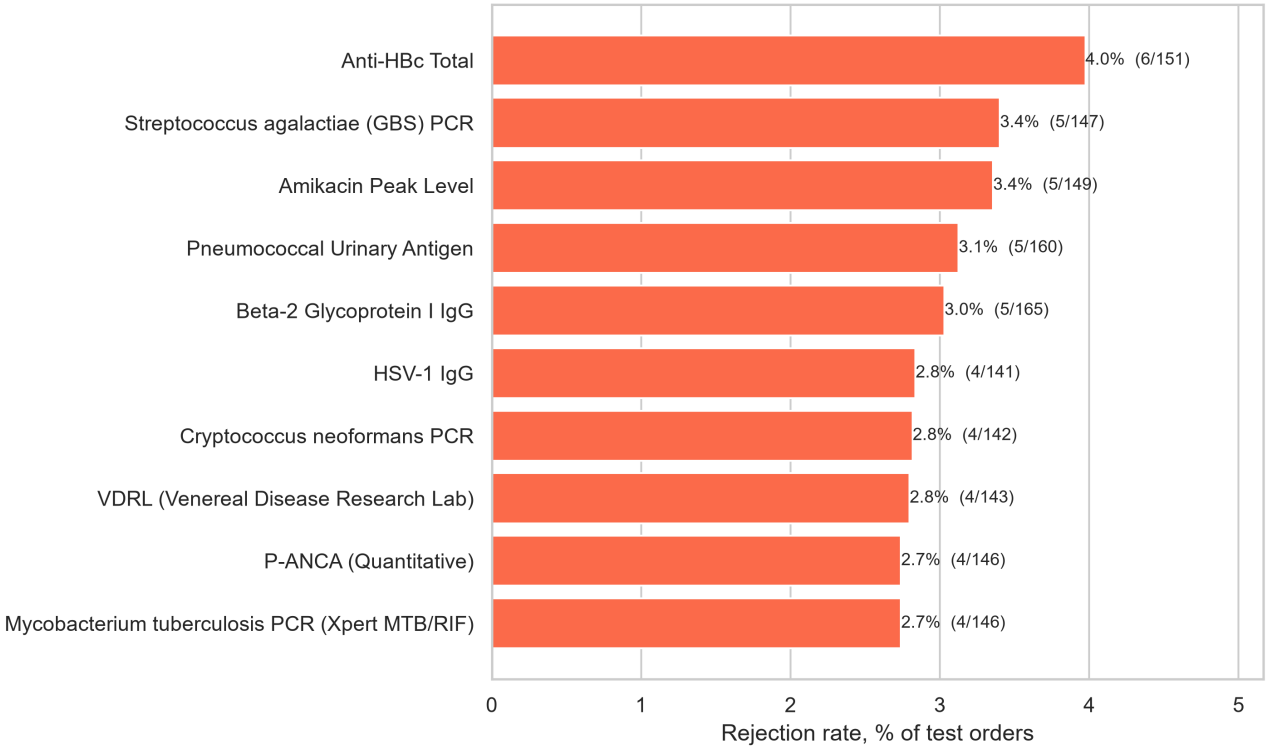
Serology also carries the largest order volume, so its marginally higher rate is also the biggest absolute contributor.

Suggested action

Reinforce collection practice (vial type, tube fill, labelling) in Serology, Immunology and Histopathology; track the rate weekly.

Top 10 Tests by Rejection Rate

The ten tests with the highest share of rejected orders - the test-level view of rejection risk.



Key insights

- Anti-HBc Total tops the list at about 4% (6 of 151 orders).
- GBS PCR and Amikacin Peak Level follow at about 3.4% each.
- The list is long-tailed: most tests reject far below these levels.

Potential bottleneck

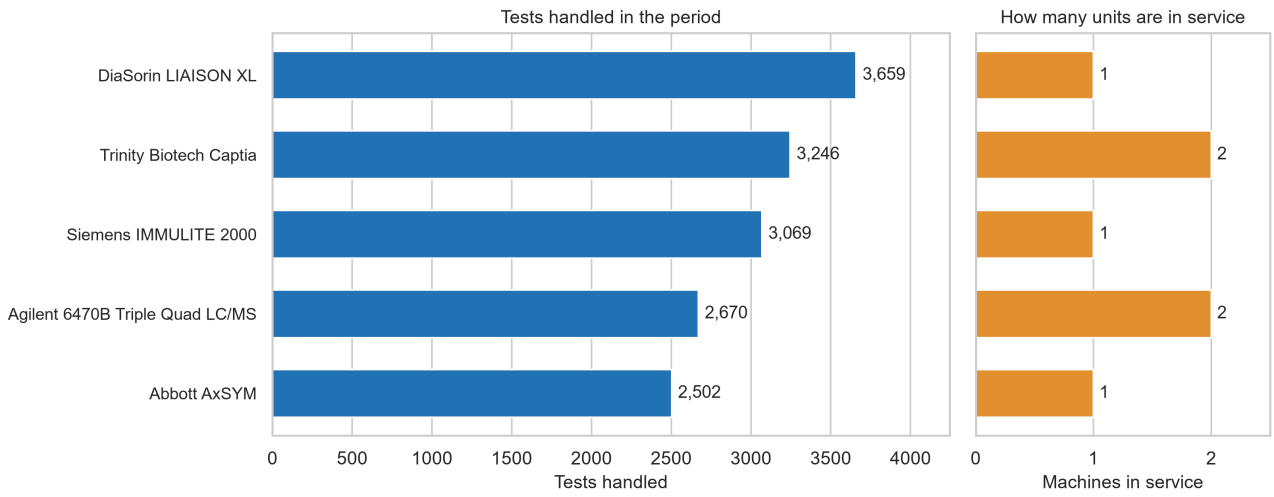
Small order counts: one or two extra rejections visibly moves these rates.

Suggested action

Check the collection protocol (sample requirements, tube handling) for the top three tests; keep the rest of the list on routine watch.

Top 5 Analyzers by Workload

The five busiest analyzers by test orders, side by side with the number of units in service.



Key insights

- DiaSorin LIAISON XL carries the most tests (3,659) - on a single machine; Siemens IMMULITE 2000 (3,069) and Abbott AxSYM (2,502) are also single-unit.
- Even the busiest needs well under one hour of rated run time per day.
- No top-5 analyzer needs more machines for today's volume.

Potential bottleneck

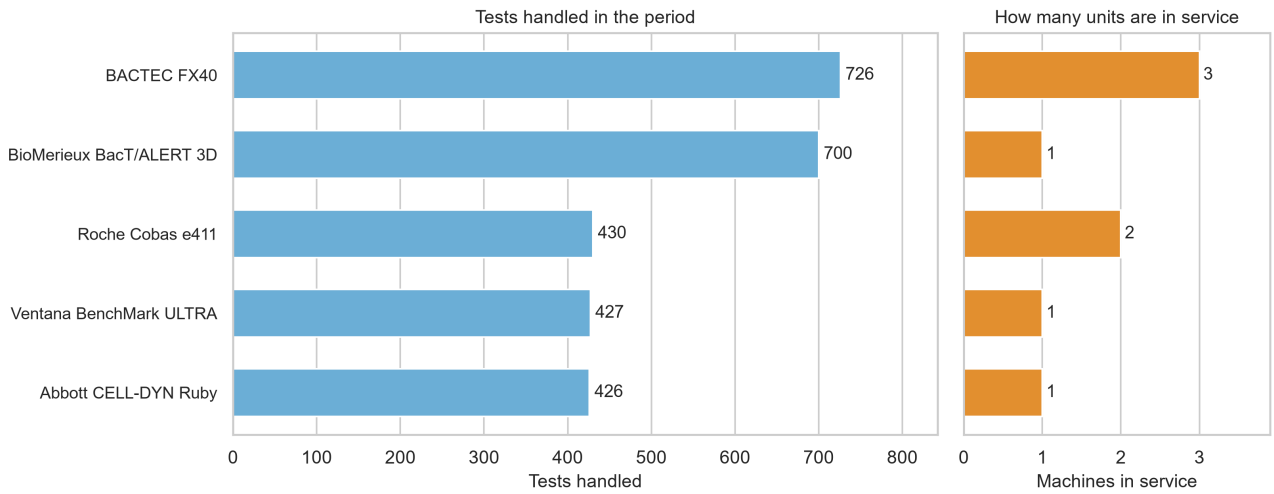
Single-unit dependence on the three heaviest analyzers - one breakdown stops roughly 26-28 tests a day each.

Suggested action

Protect the single-unit heavy hitters with preventive maintenance and a documented backup routing plan.

Bottom 5 Analyzers by Workload

The five quietest analyzers by test orders, with their machine counts - where the spare capacity sits.



Key insights

- BACTEC FX40 keeps 3 machines for 726 tests (~5 a day) - one machine would cover it in about 3 minutes a day.
- Roche Cobas e411 runs 430 tests across 2 machines (about 1 busy minute a day).
- These machines are effectively idle relative to their rated speed.

Potential bottleneck

None - this end of the scale is spare capacity, not pressure.

Suggested action

Point idle machines at QC runs, re-runs and training workload; review multi-unit fleets (e.g. BACTEC FX40) for consolidation.